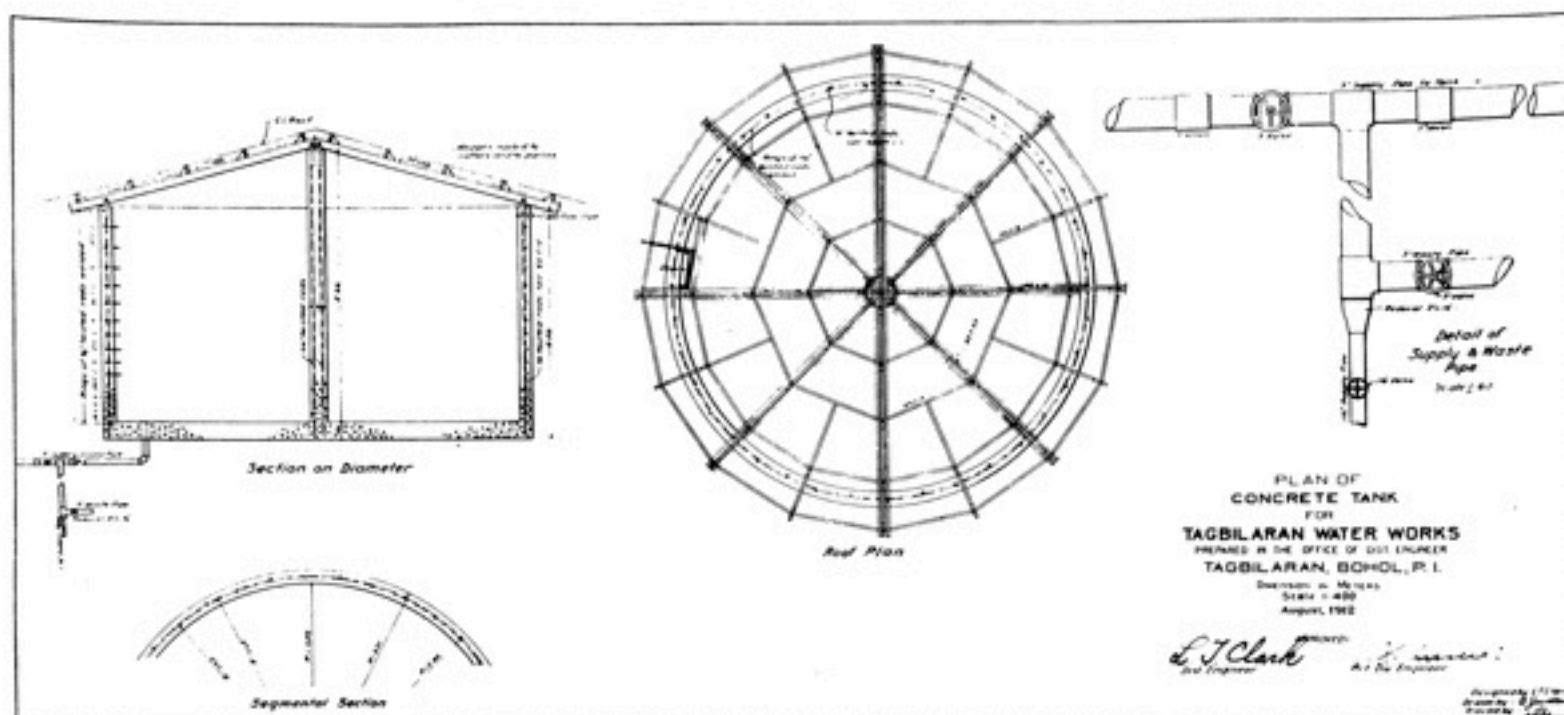




CHEMICAL ANALYSIS.

[December 28, 1910.]

Physical characters.....	Good.
Reaction.....	Alkaline.
Total solids.....	2,547.8
(a) Fixed.....	2,331.3
(b) Volatile.....	216.5
Appearance on ignition.....	Darkened slightly.
Free or saline ammonia.....	0.0096
Organic or albuminoid ammonia.....	0.0048
Oxygen consumed..... (Schulze's formula).....	0.48
Chlorine.....	1,099.5
Nitrogen as nitrates.....	Traces.
Nitrogen as nitrites.....	Nil.
Non-scale-forming ingredients.....	1,825.9
Scale-forming ingredients.....	566.9
Silica (SiO ₂).....	43.1
Oxides of iron and aluminum.....	1.2
Oxide of calcium (CaO).....	179.1
Oxide of magnesium (MgO).....	150.7
Sulphuric anhydride (SO ₃).....	297.9

BIOLOGICAL REPORT.

[November 1, 1910.]

Ciliates.....	Negative.
Pathogenic bacteria.....	Negative.
Negative for B. Coli communis.....	
Amoebae.....	Positive.
Flagellates.....	Positive.

The percentage of solids is six times that adopted by sanitary authorities for potable water and the percentage of chlorine twenty-five times, and the hardness is much greater than adopted standards. It was the opinion of the Assistant Director of Health that these conditions would better themselves after the water had flowed a sufficient time. This was found to be true and a copy of a new analysis is here given.

Water sample from well No. 192 at Tagbilaran, Bohol, No. 2.

CHEMICAL ANALYSIS.

[March 26, 1914.]

Reaction.....	Neutral.
Total solids.....	240.9
Appearance on ignition.....	No odor.
Free or saline ammonia.....	0.024
Organic or albuminoid ammonia.....	0.014
Chlorine.....	3.26
Nitrogen as nitrates.....	Traces.
Nitrogen as nitrites.....	Traces.
Silica (SiO ₂).....	97.7
Oxides of iron and aluminum.....	1.7
Oxide of calcium (CaO).....	33.7
Oxide of magnesium (MgO).....	19.99
Sulphuric anhydride (SO ₃).....	22.5

Remarks.—This water is satisfactory for drinking.

BIOLOGICAL EXAMINATION.

[March 25, 1914.]

Number of colonies per cubic centimeter.....	2,353
Presumptive test.....	Positive.
Attempt to isolate B. coli.....	Negative.
Amoebae.....	Negative.
Ciliates.....	Negative.
Flagellates.....	Negative.

The tank when first filled developed wet places on the outside due to seepage at joints. One year later the seepage had practically ceased, leaving a salt deposit on the outside of the tank. This deposit is due to the amount of solids in the water.

This tank has since been painted a cream color with "Stonetex," recommended by the Trus Con Laboratories; seepage having completely ceased, the inside of the tank required no further waterproofing. It was contemplated that it would require waterproofing and a trial of Trus Con waterproofing paste was to have been made.

The painting with "Stonetex" has not been entirely satisfactory, but the appearance of the tank is improved to a certain extent, and by the addition of another coat of paint the white deposits may be made invisible. The deposits were scraped off before the paint was applied.

Map of Tagbilaran appears herewith showing complete pipe line and all public hydrants. The main line from well to tank is 3 inches in diameter with branch lines 1½ inches in diameter. Distance from well to tank is 860 meters. Lift is from 12.50 to 16.50 meters, the intake being in the bottom of the tank.

Water is pumped by a Fairbanks-Morse kerosene pumping engine, 8 horsepower, with a 4-inch intake and a 3-inch discharge. Cost of engine, ₱1,950.29.

At first the use of this machine was not very successful, due to lack of experience with this kind of an internal-combustion engine. However, during the last nine months that this pumping plant was operated, very little trouble has been encountered in the operation of the engine.

Total cost of the installation of pipes and tank as shown by the drawing appearing herewith was ₱12,195.07. This does not include cost of pumping plants which are listed separately. There are now in use 9 public and 65 private hydrants, all with ½-inch compression bibbs. There are also 4 other hydrants for official use.

At one time, the artesian well became bridged over to such an extent that the water would not rise fast enough for the pump. It was found necessary to have it cleaned out and a pocket drilled below the water-bearing stratum. This was done by one of the Bureau